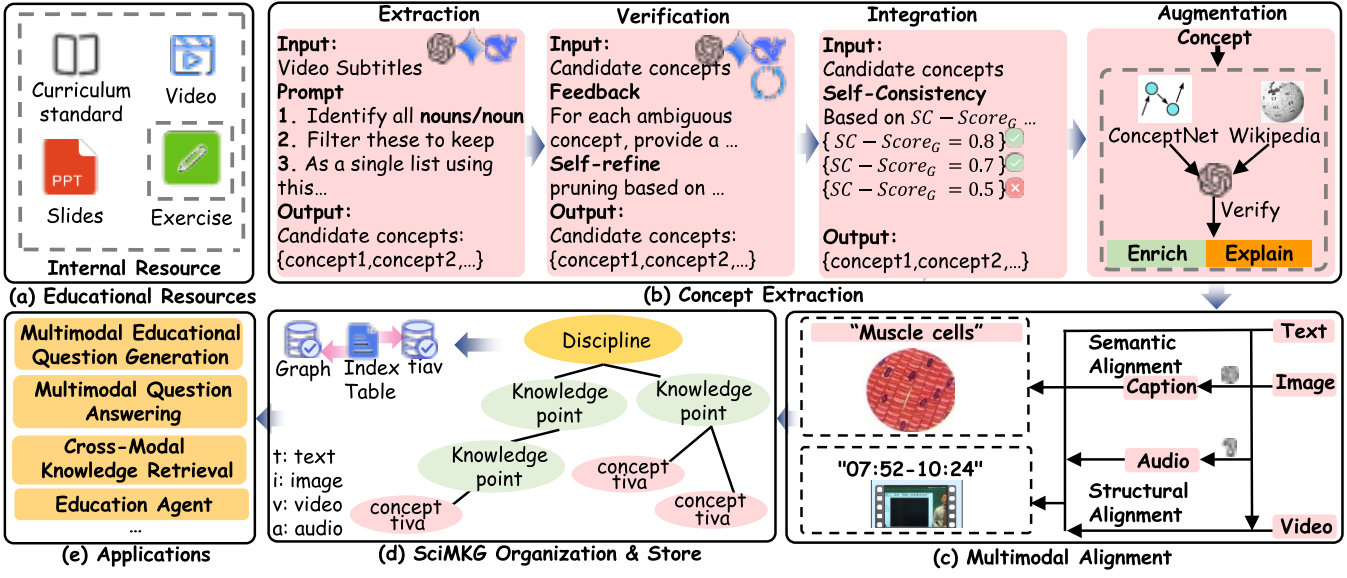

Curriculum standard
Video
PPT
Slides
Exercise
Internal Resource
(a) Educational Resources
Extraction
Input:
Video Subtitles
Prompt
1. Identify all nouns/noun
2. Filter these to keep
3. As a single list using this...
Output:
Candidate concepts:
{concept1,concept2,...}
Verification
Input:
Candidate concepts
Feedback
For each ambiguous concept, provide a ...
Self-refine
pruning based on ...
Output:
Candidate concepts:
{concept1,concept2,...}
Integration
Input:
Candidate concepts
Self-Consistency
Based on $SC-Score_G$ ...
$\{SC-Score_G=0.8\}$
$\{SC-Score_G=0.7\}$
$\{SC-Score_G=0.5\}$
Output:
{concept1,concept2,...}
Augmentation
Concept
ConceptNet
Wikipedia
Verify
Enrich
Explain
(b) Concept Extraction
Multimodal Educational Question Generation
Multimodal Question Answering
Cross-Modal Knowledge Retrieval
Education Agent
...
(e) Applications
Graph
Index tiav Table
Discipline
Knowledge point
Knowledge point
Knowledge point
concept tiva
concept tiva
concept tiva
t: text
i: image
v: video
a: audio
(d) SciMKG Organization & Store
"Muscle cells"
"07:52-10:24"
Semantic Alignment
Caption
Audio
Structural Alignment
Text
Image
Video
(c) Multimodal Alignment